Julia Meshcheryakova

ML Engineer / Data Scientist, 15+ years in IT

ML Engineer with an activist mindset, focusing on projects that improve lives by means of technology and innovation. Starting in 2011 as a full-stack software engineer, I quickly progressed to a Team Lead position. With my strong mathematical and algorithmic background and technical expertise, in 2019, I transitioned into Machine Learning and Data Science. Totals: 20+ completed projects, award-winning innovative solutions with significant commercial and social effect. I'm most inspired by problems that may not have a known solution yet, like the orthodontic CAD system I built. Today I build solutions around models and run the experiments and evaluations on them; next, I want to move from that into research on the models themselves. UK Global Talent Visa holder (on ILR now).

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WORK EXPERIENCE

Senior ML Engineer & Data Scientist, Frontier Company (prev. ISE — Industry Solutions Engineering) — Microsoft 10/2023 – Present · London, UK

- Core focus: agentic research workflows, knowledge extraction from variable unstructured data formats, evaluation without reliable ground truth, guardrails implementation, model fine-tuning. My black-box comparison feature for external models was later adopted into the organisation-wide data-science evaluation framework.
- Security and Responsible AI: full risk mapping and assessments; customer workshops and upskilling.
- Technical lead across 8 engagements with key Microsoft customers in pharma, healthcare, industrial manufacturing and market research — from problem framing and research with domain experts to production and handover together with the customer engineering team.
- Open source & product feedback loop: governance-group member for the RAG Experiment Accelerator; converted engagement findings into Azure product-group feature requests and playbooks.

Senior ML Engineer, Member of AI Technical Board and Tech Forum — DataArt (global software engineering firm) 02/2019 – 09/2023 · Voronezh, Russia → Yerevan, Armenia → London, UK

- Led ML teams with 7-10 members, responsible for the full development cycle: from the research and solution design to deployment to clouds.
- 10+ completed projects of different sizes, top-3 performer of AI/ML Center of Excellence in 2022: working on business accelerators, interviewing candidates, mentoring, and reviewing trainees.
- Acknowledged contributor to DataArt's 2020 innovation award from Corporate Vision and Employer Brand, and to its recognition in the Gartner® Hype Cycle™ for Life Science Clinical Development, 2022.

Senior .Net Developer, Team / Tech Lead — DataArt 12/2011 – 01/2019 · Voronezh, Russia

- Led distributed teams with 10+ people within product-led companies in healthcare, travel, and finance, from start-ups to big enterprises (6 projects). Solved complicated architectural challenges, performed task distribution, code review, and junior team members mentoring.
- 6 completed projects from start-up size to big enterprise companies.

Trainee C Developer (RELEKS) & C++ Teaching Assistant (Voronezh State University) 2010 – 2014 · Voronezh, Russia

- Trainee C developer — conducted research on a memory management algorithm used in DBMS LINTER. Devised approach ~16% more efficient.
- 2y of teaching "Programming in C++" at Voronezh State University.

SELECTED PROJECTS

Agentic framework for a National Healthcare Provider 11/2025 – 05/2026

- Technical lead: designed and delivered an agentic framework for a highly regulated healthcare environment — combining agentic capabilities with strict guardrails, and built to scale across the organisation.
- Led the Responsible AI assessment and full risk mapping; resolved a stalling multi-party engagement by resetting it around a use case all parties could deliver.

Entity extraction & evaluation for reported-case processing, Global Pharmaceutical Company 05/2026 – 09/2026

- Built the experimentation and evaluation pipeline for LLM-based entity extraction from noisy, inconsistently-labelled real-world case reports.
- Designed a strategy for mutually dependent fields — combining them, pre-processing the ground truth and post-processing at the evaluation step — raising extraction accuracy 65% → 86% and cutting token cost ~3×.
- Delivered a production-ready MVP with metrics monitoring, deployed as an Azure App Service.

Agentic oncogenomics research flow for Global Pharmaceutical Company 09/2026

- **Award:** 3rd place, Microsoft Global Hackathon 2026 (Executive Challenge for Co-Innovation with Customers); winning project of the Healthcare & Government studio.
- Designed the agent orchestration for the research flow on top of Microsoft Discovery.
- Aligned the design with computational biologists — mapping their workflows, methods, pain points and expectations onto agent roles and hand-offs.

GraphRAG-powered research assistant for Global Pharmaceutical Company 03/2024 – 09/2025

- Led the development of Neo4j-based Graph RAG solution turning PubMed into explainable, source-linked insights.
- Developed evaluation criteria for the generated insights with no ground truth available, and operationalised the evaluation practice across the engagement.
- Upskilled the customer, providing series of training sessions during the engagement and handover; led the RAI assessment of the solution.

Reproducible-research tool for Global Pharmaceutical Company 03/2026

- **Award:** winning project, internal Microsoft Healthcare studio hackathon; presented to customer leadership and fed into a new engagement track.
- Owned the visualisation track: built auditable context sharing between MCP tools and a visualisation tool, and co-developed an integrated dashboard with the customer.
- Proposed and implemented “Git for researchers” — a reproducible research flow with checkpoints to recover and share experiment context. An unassigned idea that became the app’s headline feature.

Parsing receipts with LLM for Global Market Research Leader 09/2024 – 03/2025

- **Award:** Internal engineering excellence award, FY25 Q4.
- Set up the LLM fine-tuning, evaluation and monitoring pipelines (including custom task-related metrics).
- Raised extraction accuracy 76% → 91% and cut manual effort by 42%, enabling scale across countries, channels and document formats.
- Integrated the fine-tuned model into the customer's existing ecosystem and ran the security assessment of the full solution for LLM-connected threats.
- Devised security assessment of the full customer solution focusing on LLM-connected threats.

Orthodontic CAD system for automating lingual aligners design 02/2019 – 01/2020

- Developed the algorithms for automatic 3D surface segmentation: locating points/areas of interest and topological features (handles), hole filling, and completion of enclosed surfaces — to remove the main bottleneck for scaling.
- Created detailed documentation of algorithms for patents registration.
- Dental Products Report included the app in the top-5 software products of 2020; in December 2021, company received 510(k) FDA clearance for the app.
- The company grew to 5000+ patients treated in 1y after the automated version release.

TECHNICAL SKILLS

Top Skills: Machine Learning, Data Science, Software Engineering, Problem framing & creative problem-solving, Team/Tech Leadership, Public Speaking, Mentorship, .Net

Data-science and ML: Agentic systems & multi-agent orchestration, LLM evaluation without ground truth, RAG & GraphRAG, Fine-tuning and LoRA, Transformer architectures, LLM, NLP, Classical ML, Neural networks (incl. RNNs), Computer Vision

Programming and Markup Languages: Python, T-SQL, JS, TypeScript, C++, HTML, XML, CSS, C, C#

Technologies and Frameworks: Azure OpenAI, Azure AI Foundry, Azure ML, Prompt flow, MLflow, Neo4j, Microsoft Discovery, Docker, PyTorch, TensorFlow, NumPy, Pandas, SciPy, scikit-learn, spaCy, OpenCV, Google OR-Tools, Angular, jQuery, MS .NET Framework and .NET Core, ASP.NET MVC and Core

Clouds: MS Azure, AWS

Databases: MS SQL Server, MongoDB, Redis, Snowflake

EDUCATION

Postgraduate, Master's and Bachelor's degree in Applied Mathematics and CS
Voronezh State University (Russia) · 2007 – 2016

COURSES

AI Performance Engineering (4-month course) 03/2026 – 06/2026
Nebius Academy · LLM architectures and neural networks (transformers, fine-tuning and LoRA), AI agents, MLOps, performance engineering (speculative decoding, quantization, CUDA kernels), and LLM post-training, from SFT to RL.

LANGUAGES

English: Full Professional Proficiency (IELTS 8.5, 2026)
Russian: Native

CONFERENCES & TALKS

Security-insights forum — internal, Microsoft 2025
Authored and presented an LLM-threat security assessment (actors, threats, mitigations) for a customer's production environment.

Security Day workshop: Cutting Corners Hero — internal, Microsoft 2026
Built and delivered a gamified workshop on secure-by-design trade-offs for LLM systems at a Security Day across the Frontier Company organisation; strong feedback, accepted for internal security training and being scaled across teams.

Data Science Europe 11/2022
Presented DataArt AI&ML CoE expertise. Attracted dozens of candidates and several clients.

COMMUNITY & MENTORING

LLMs-from-scratch study group lead & Dev Academy mentor — Microsoft (internal) 2024 – 2026
Organised and led a 17-engineer study group building LLMs from scratch. Mentor in Microsoft's Dev Academy: 6 sessions across the Engineer and Data Science tracks (~30 engineers Microsoft-wide) and 2 hands-on AI Learning Day labs.

Top Active Community Member — StackOverflow 05/2013 – Present
~180k people reached, 70 badges, 3200+ reputation.

Mentor in Machine Learning — TechLeaders (mentors program) 09/2020 – 12/2020 · Voronezh, Russia
Created an experimental program for Machine Learning that was then accepted as ML internship program for graduates at DataArt.

Ambassador — IT-Cup (intellectual games tournament in the IT industry) 09/2012 – 09/2019 · Voronezh, Russia
Organised the tournament in 2012, 2013, 2014, 2015, 2016, 2017, 2019.

PERSONAL PROJECTS

Founder & sole engineer, Quizrun (quizrun.org): A hybrid sport I co-invented: a running event where lateral-thinking puzzles solved en route subtract time from your finish. 500+ participants from 60+ cities in 20+ countries so far. Built solo, agent-assisted, applying production engineering practice from the first commit: React/TypeScript + Vite, Supabase/Postgres with row-level security and role-based access, CI with gated database migrations, staging and production on Cloudflare. Covers registration, scoring, track analysis, messaging and an analytics dashboard for organisers.

Food Enough (food-enough.pages.dev): Your lens on any menu: scan it in any language, see what you can actually eat, then get help choosing. It grew from my own pain as a picky eater with choice paralysis. In development, shipping soon.

CogniRun: Extending the Quizrun engine into a mass-participation platform — turning run-and-solve data into population-level insight on how people reason under physical load. In active development.

PUBLICATIONS

Superbacteria: The War We're Going to Lose Without AI and Big Data
Julia Meshcheryakova, Denis Tomilin · August 16, 2023 · healthcarebusinesstoday.com

Holiday Season: Predicting and Tolerating High Load in Business - Dealing With Unexpected Anomalies
Julia Meshcheryakova · January 11, 2023 · hackernoon.com

The Relevance of Cloud-Based Technologies in Modern Manufacturing
Julia Meshcheryakova, Max Ivannikov · October 20, 2022 · iotforall.com — top search result in google on topic. Published in computer-automation.de as "Die Bedeutung von Cloud-basierten Technologien in der modernen Fertigung".

VOLUNTEER EXPERIENCE

IT-support, coordinator, full-stack contributor — eco-dvor & eco-taxi (vrn) & etonemusor.ru 03/2020 – 03/2023 · Voronezh, Russia → Yerevan, Armenia
Non-commercial projects, providing citizens with a way to recycle their waste and educating them.

- Automated coordinators' workflows for route planning and stats, cutting single-run time by 30%.
- Integrated external data from 5+ eco-projects, adding 100+ points to etonemusor.ru map. Delivered 10+ feature proposals and built an e-map with mobile points and contacts.
- Created auto-generated 1-page waste classification memo and authored 20+ educational posts.
- Scaled project from 15 → 50 mobile points; the ecodvor community reached 5k, and finally, the commercial recycling company launched in Voronezh.